Learned maintenance content has bounded physical utility

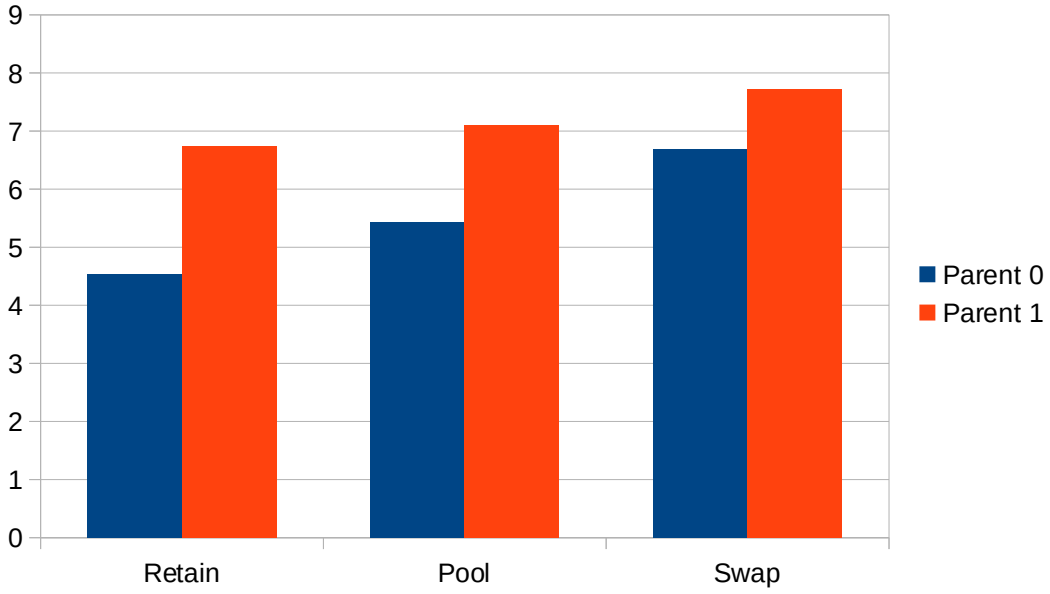
Externally supplied material CA • central budget • two parent bodies

Cue (8 steps) → gap (2) → mechanical contact (4)

Incident strain teaches delta; purchased reserve absorbs demand before injury.

A Acquired content, stable relation

Tissue loss (mass units) • +96 steps; writes off



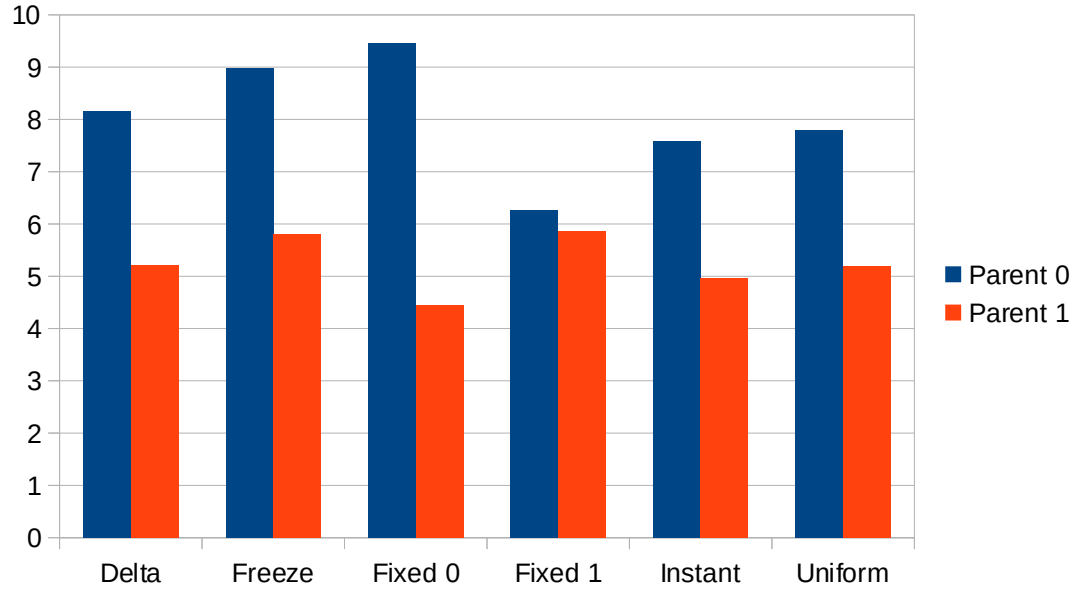
Retain vs pool: 16.6% / 5.1% less loss.

Same synthesis; total expenditure differs.

Retain, pool or swap the acquired readout.

B Unannounced relation change

Tissue loss (mass units) • steps 225–320



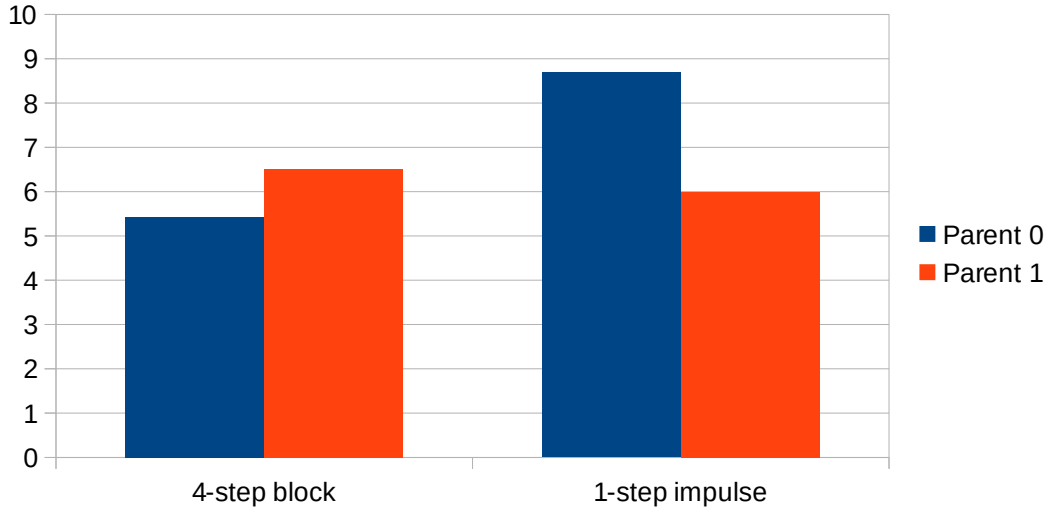
Delta vs freeze: 9.17% / 10.24% less loss.

Delta vs instant: 7.51% / 4.83% more loss.

Both fixed maps retained; no post-hoc selection.

C Stable relation; acquired content frozen

Loss reduction vs instant (%) • +96 steps



Block: 5.42 / 6.51%; impulse: 8.69 / 5.98%.

Same external strain integral; body demand can differ.

Body trade-offs: delta minus instant

After relation change	Parent 0	Parent 1
Subsequent uptake	+0.109	−0.388
Terminal mass	+0.340	−0.258
Terminal energy	−0.718	+0.250

No consistent net physical advantage.

In C, uptake and terminal mass/energy are modestly higher in all four paired conditions.

Impulse minus block: extra absolute loss avoided +0.271 / +0.028 mass units (parents 0 / 1).

Stable-relation benefit ≠ reversal success.

Matched synthesis ≠ matched total cost; simple controls pay unused predictor-capacity fees.

Registered old-cohort death ≠ cumulative births ≠ retained newborn. Two parents; correlated forks.

Ordinary delta; observable incident-strain label. No new predictor or general net advantage.